Physical Design Review: Digital CIM

Yunqi's Children

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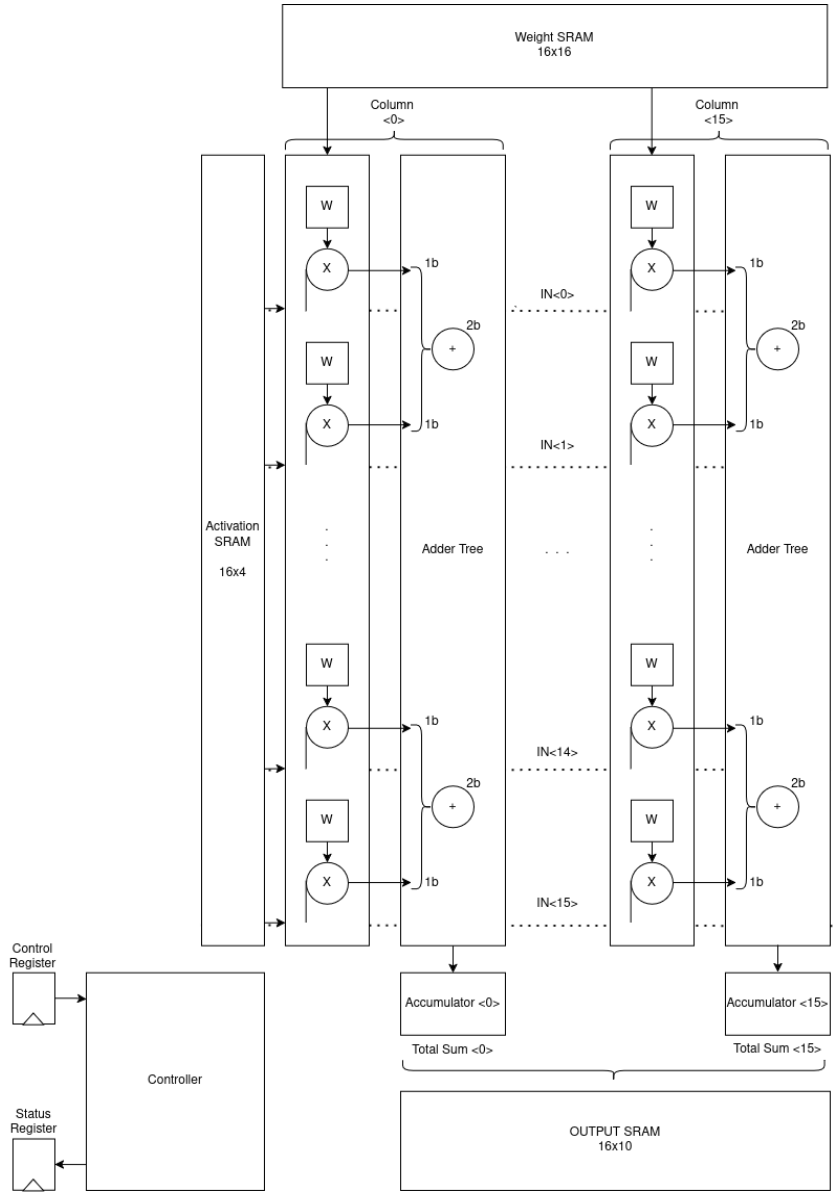
System Specification Recap

- Design and verify an all-digital CIM device for convolutional neural network MAC operations
- 16x16 core
- 1-bit weights and 4-bit activations
- Beginning target: 50-100 MHz



System Specification Recap

- Fully synthesized latch-based CIM core
  - No custom layout
  - Standard cell wrapper

```
// Latch Wrapper Module
module dcim_latch_cell (
    input  wire d_in,   // input data
    input  wire wl_en,  // write enable
    input  wire rst_n,  // reset
    output wire q_out   // output data
);

    LHCNQD1 u_latch (
        .D(d_in),
        .E(wl_en),
        .CDN(rst_n),
        .Q(q_out)
    );

endmodule
```

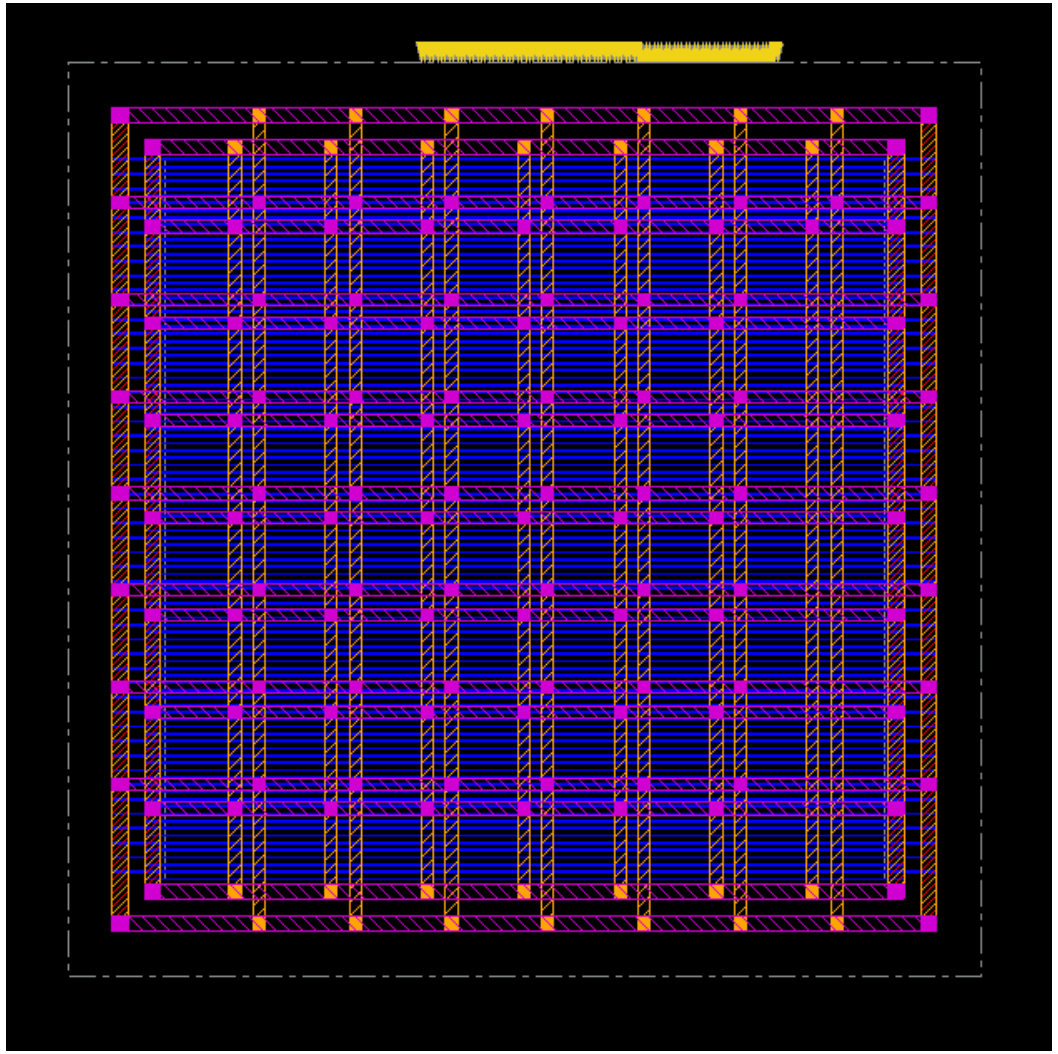
System Specification Recap

	Synthesis Specs	Post-Apr Specs
Clock	500 MHz	500 MHz
Power	5.393 mW	19.37 mW
Area	19,163 um^2	51,437 um^2

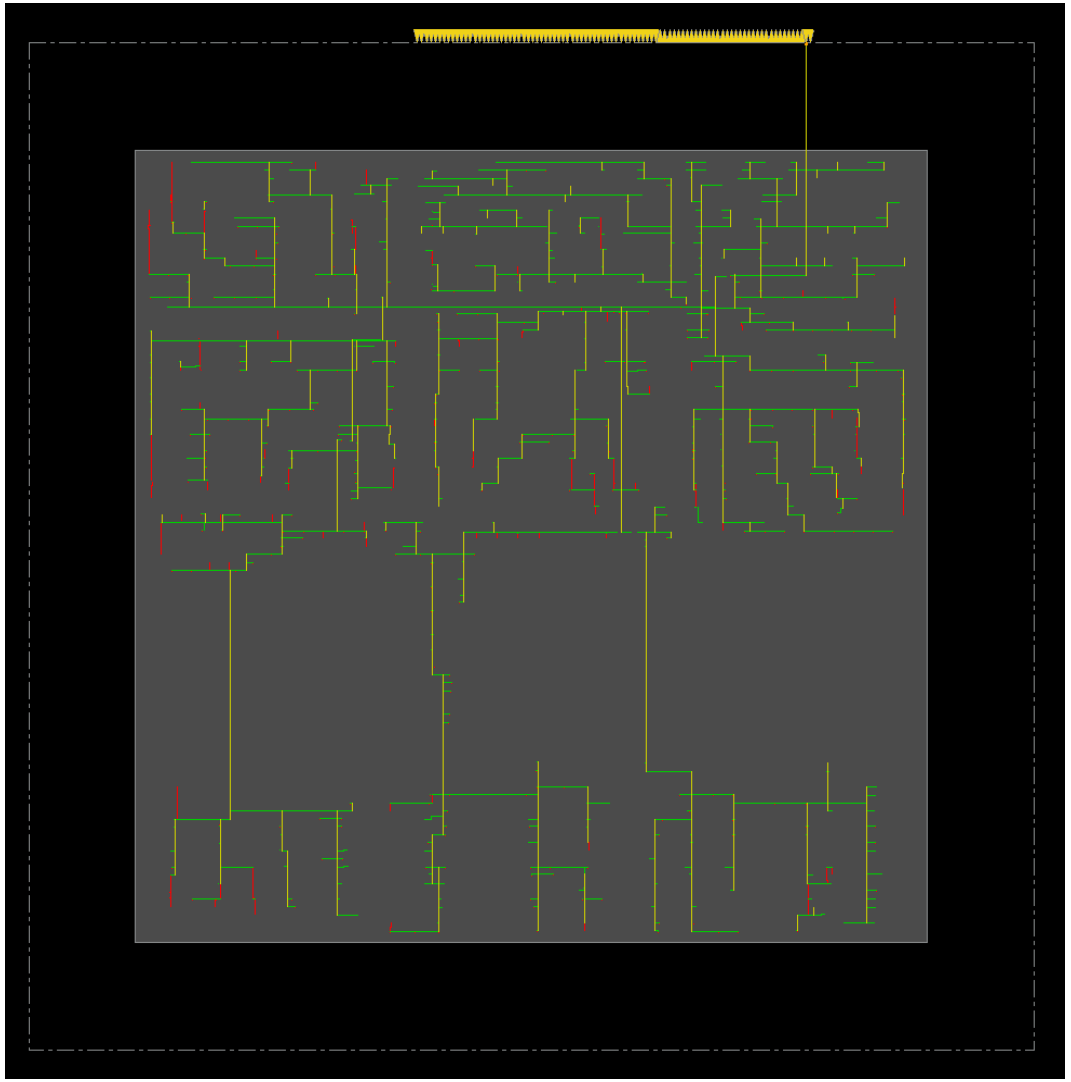
Cell Utilization

```
==============================
Floorplan/Placement Information
==============================
Total area of Standard cells: 31969.080 um^2
Total area of Standard cells(Subtracting Physical Cells): 29774.160 um^2
Total area of Macros: 0.000 um^2
Total area of Blockages: 0.000 um^2
Total area of Pad cells: 0.000 um^2
Total area of Core: 31969.080 um^2
Total area of Chip: 51437.880 um^2
Effective Utilization: 1.0000e+00
Number of Cell Rows: 99
% Pure Gate Density #1 (Subtracting BLOCKAGES): 100.000%
% Pure Gate Density #2 (Subtracting BLOCKAGES and Physical Cells): 93.134%
% Pure Gate Density #3 (Subtracting MACROS): 100.000%
% Pure Gate Density #4 (Subtracting MACROS and Physical Cells): 93.134%
% Pure Gate Density #5 (Subtracting MACROS and BLOCKAGES): 100.000%
% Pure Gate Density #6 (Subtracting MACROS and BLOCKAGES for insts are not placed): 93.134%
% Core Density (Counting Std Cells and MACROs): 100.000%
% Core Density #2(Subtracting Physical Cells): 93.134%
% Chip Density (Counting Std Cells and MACROs and IOs): 62.151%
% Chip Density #2(Subtracting Physical Cells): 57.884%
```

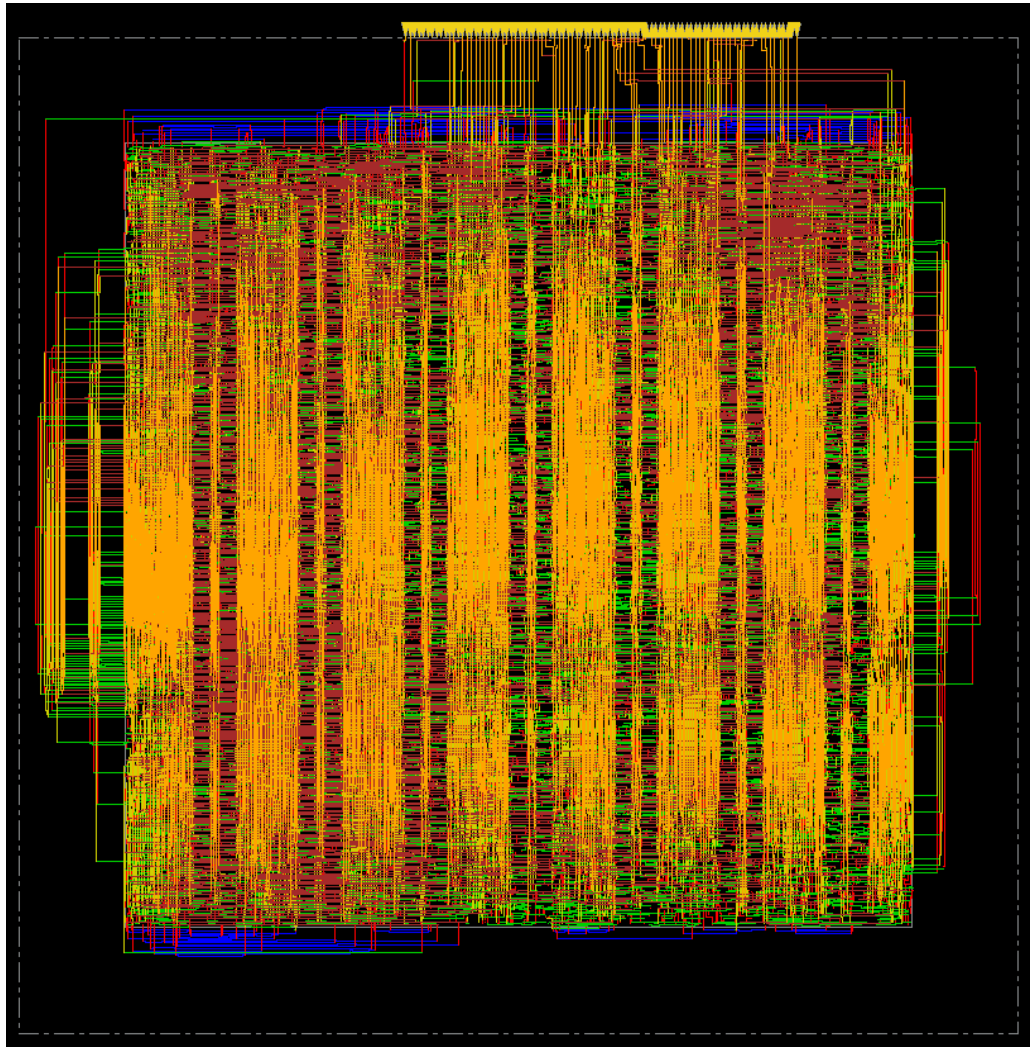
Power Grids



Clock Tree



Signal Routing



Power Estimates

```
Total Power
-----------------------------------------------------------
Total Internal Power:        13.03517048        67.2792%
Total Switching Power:        6.32732491        32.6576%
Total Leakage Power:          0.01224439         0.0632%
Total Power:                 19.37473978
-----------------------------------------------------------
```

Full Hold/Setup Reports

```
------------------------------------------------------------------
     optDesign Final SI Timing Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.752  |  0.752  |  1.005  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   852   |   820   |   76    |
+--------------------+---------+---------+---------+

+----------------+-----------------------------+------------------+
|                |              Real           |       Total      |
|    DRVs        +-----------------+-----------+------------------|
|                | Nr nets(terms)  | Worst Vio | Nr nets(terms)   |
+----------------+-----------------+-----------+------------------+
|   max_cap      |      0 (0)      |   0.000   |      0 (0)       |
|   max_tran     |      0 (0)      |   0.000   |      0 (0)       |
|   max_fanout   |      0 (0)      |     0     |     11 (11)      |
|   max_length   |      0 (0)      |     0     |      0 (0)       |
+----------------+-----------------+-----------+------------------+

Density: 93.134%
------------------------------------------------------------------
```

```
------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.205  |  0.205  |  0.289  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   852   |   820   |   76    |
+--------------------+---------+---------+---------+

Density: 93.134%
------------------------------------------------------------------
```

- Set-up Time Report
- Hold time Report

Critical Hold/Setup Paths

```
Path 1: MET Hold Check with Pin core_cim_wt_data_q_reg_7_0/CP
Endpoint:   core_cim_wt_data_q_reg_7_0/D              (v) checked with  leading
edge of 'clk'
Beginpoint: u_digital_cim/u_controller/state_reg_2_/Q (^) triggered by  leading
edge of 'clk'
Path Groups: {clk}
Analysis View: typical_analysis_view
Other End Arrival Time          0.015
+ Hold                          0.033
+ Phase Shift                   0.000
- CPPR Adjustment               0.000
+ Uncertainty                   0.200
= Required Time                 0.248
  Arrival Time                  0.453
  Slack Time                    0.205
     Clock Rise Edge                 0.000
     + Clock Network Latency (Prop)  0.157
     = Beginpoint Arrival Time       0.157
```

- Critical Path: Hold time

```
Path 1: MET Setup Check with Pin u_digital_cim/say_less_9__u_accumulator/acc_
reg_reg_9_/CP
Endpoint:   u_digital_cim/say_less_9__u_accumulator/acc_reg_reg_9_/D (^)
checked with  leading edge of 'clk'
Beginpoint: u_act_sram/rdata_reg_1_/Q                                (^)
triggered by  leading edge of 'clk'
Path Groups: {clk}
Analysis View: typical_analysis_view
Other End Arrival Time          0.157
- Setup                         0.008
+ Phase Shift                   2.000
+ CPPR Adjustment               0.000
- Uncertainty                   0.200
= Required Time                 1.950
- Arrival Time                  1.198
= Slack Time                    0.752
     Clock Rise Edge                 0.000
     + Clock Network Latency (Prop)  0.015
     = Beginpoint Arrival Time       0.015
```

- Critical Path: Setup time

Clock skew delay

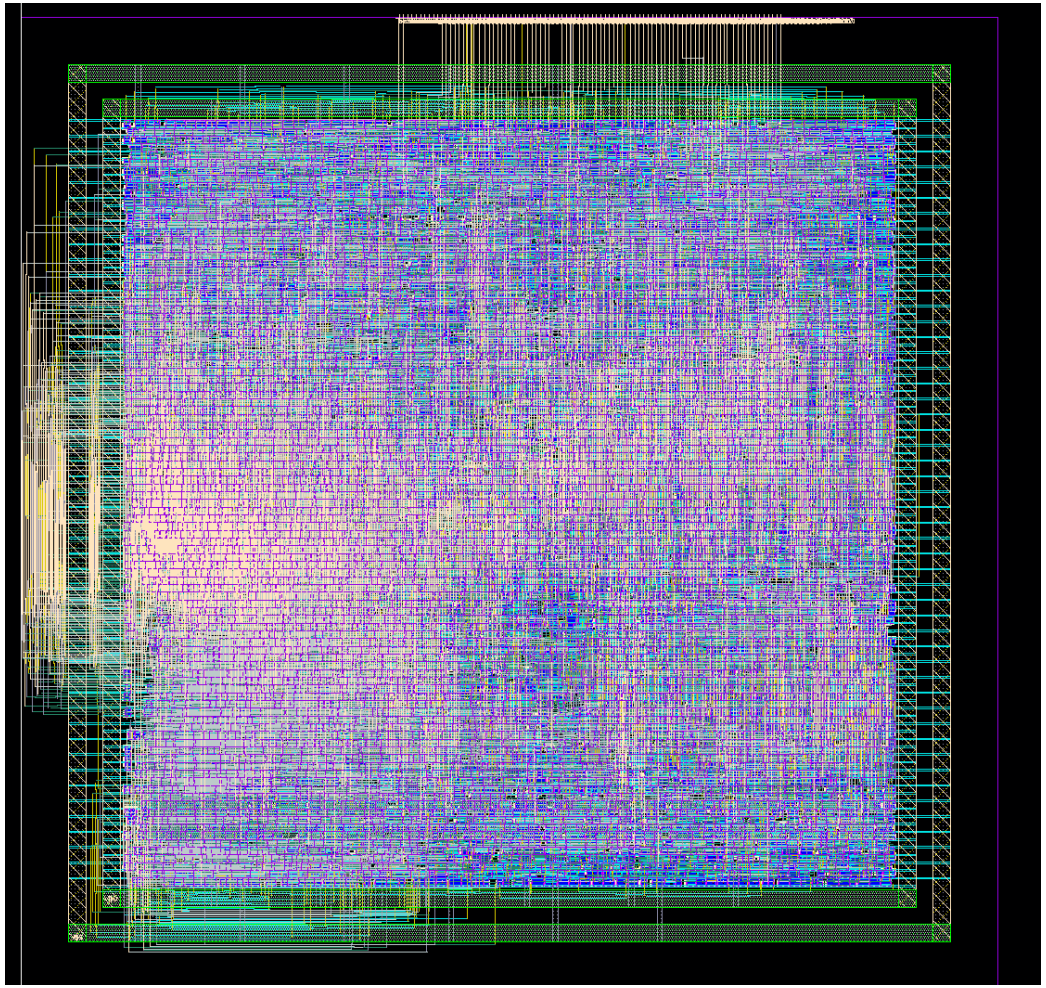
```
Timing for timing corner typical_delay_corner:both.early, min clock_path:
==========================================================================

PathID    : 1
Path type : skew group clk/my_constraint_mode (path 1 of 1)
Start     : clk
End       : u_out_sram/entry_reg_5__0_/CP
Delay     : 0.097
```

```
Timing for timing corner typical_delay_corner:both.early, max clock_path:
==========================================================================

PathID    : 2
Path type : skew group clk/my_constraint_mode (path 1 of 1)
Start     : clk
End       : u_weight_sram/entry_reg_15__6_/CP
Delay     : 0.116
```

Full Module Layout



DRC and LVS Status

- DRC "clean"

Check / Cell	Results	Flat
Check DOD.R.1	1	1
Check PO.DN.2	2	2
Check PO.R.8	136	4227
Check DPO.R.1	1	1
Check M5.DN.1	1	1
Check M6.DN.1	2	2
Check M7.DN.1	1	1
Check M8.DN.1	1	1
Check M9.DN.1:L	1	1
Check M9.DN.2:L	1	1
Check CSR.R.1:NWi	42	42
Check CSR.R.1:PPi	122	122
Check CSR.R.1:NPi	122	122
Check CSR.R.1:COi	1000	1000
Check CSR.R.1:M1i	1000	1000
Check CSR.R.1:M1_real	1000	1000
Check CSR.R.1:M2i	1000	1000
Check CSR.R.1:M2_real	1000	1000
Check CSR.R.1:M3i	1000	1000
Check CSR.R.1:M3_real	1000	1000
Check CSR.R.1:M4i	573	573
Check CSR.R.1:M4_real	573	573
Check CSR.R.1:M5i	312	312
Check CSR.R.1:M5_real	312	312
Check CSR.R.1:M6i	127	127
Check CSR.R.1:M6_real	127	127
Check CSR.R.1:M7i	8	8
Check CSR.R.1:M7_real	8	8
Check CSR.R.1:VIA1i	1000	1000
Check CSR.R.1:VIA2i	1000	1000
Check CSR.R.1:VIA3i	1000	1000
Check CSR.R.1:VIA4i	1000	1000
Check CSR.R.1:VIA5i	1000	1000
Check CSR.R.1:VIA6i	1000	1000
Check CSR.R.1:ODi	1000	1000
Check CSR.R.1:POi	1000	1000
Check DM1.R.1	1	1
Check DM2.R.1	1	1
Check DM3.R.1	1	1
Check DM4.R.1	1	1
Check DM5.R.1	1	1
Check DM6.R.1	1	1
Check DM7.R.1	1	1
Check DM8.R.1	1	1
Check DM9.R.1	1	1
Check ESD.WARN.1	1	1

DRC and LVS Status

- LVS clean
- LVS warnings:
  - Unbalanced smashed mosfets were matched
  - Ambiguity points were found and resolved arbitrarily

chip_top.lvs.report

```
                ##############################################
                ##                                          ##
                ##        C A L I B R E   S Y S T E M       ##
                ##                                          ##
                ##            L V S   R E P O R T           ##
                ##                                          ##
                ##############################################



REPORT FILE NAME:         chip_top.lvs.report
LAYOUT NAME:              svdb/chip_top.sp ('chip_top')
SOURCE NAME:              /afs/umich.edu/group/e/ece598to/w26/group1/digital_cim/v2lvs/chip_top.cdl ('chip_top')
RULE FILE:                _calibre.lvs_
CREATION TIME:            Tue Mar 17 16:07:28 2026
CURRENT DIRECTORY:        /afs/umich.edu/group/e/ece598to/w26/group1/digital_cim/final/Calibre/LVS
USER NAME:                paaditya
CALIBRE VERSION:          v2025.3_28.17    Fri Aug 1 15:10:27 PDT 2025



                             OVERALL COMPARISON RESULTS



                       #      ##################       _   _
                      #       #                #      *   *
                  #   #       #     CORRECT    #        |
                   # #        #                #       \___/
                    #         ##################


  Warning:  Unbalanced smashed mosfets were matched.
  Warning:  Ambiguity points were found and resolved arbitrarily.


*****************************************************************************************************
                                      CELL  SUMMARY
*****************************************************************************************************

  Result        Layout                      Source
  -----------   -----------                 --------------
  CORRECT       chip_top                    chip_top



*****************************************************************************************************
                                      LVS PARAMETERS
*****************************************************************************************************
```

Physical Verification & Tapeout Readiness

	Readiness
✅	DRC/LVS
✅	Timing
✅	Power/area
✅	Clocking
	Post-APR Verification

Next Steps

- Todo: Post-APR verification and Integration
- Week 13: Tapeout review (DRC/LVS clean)
- Week 14: Trial Tapeout
- Week 15: Final Tapeout